PRML Assignment 2: Classification Experiment Report

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Abstract

This report investigates the classification performance of several machine learning models on a 3D dataset generated using a modified "make-moons" algorithm. The study compares Decision Trees, AdaBoost (with Decision Trees), and Support Vector Machines (SVM) across three distinct kernel functions: Linear, Polynomial, and RBF. Experiments demonstrate that for highly nonlinear 3D manifold data, kernel-based methods and ensemble learning significantly outperform linear boundaries. AdaBoost achieved the highest accuracy of 0.990, followed closely by RBF SVM at 0.988, whereas Linear SVM remained inadequate with an accuracy of 0.698.

1. Introduction

Classification is a core task in pattern recognition, aiming to assign input vectors into distinct categories. In realistic datasets, classes are often intertwined in high-dimensional space, requiring non-linear decision boundaries.

In this assignment, we work with a 3D dataset (1000 training points and 500 test points) where two classes (C0 and C1) are generated sequentially to form a curvilinear structure. The task involves:

(1) Evaluate the baseline classification performance of Decision Trees.
(2) Analyze the performance boost provided by ensemble methods like AdaBoost.
(3) Investigate the impact of different SVM kernels on capture-ability of the decision surface.

2. Methodology

2.1 Classification Models

Decision Trees (DT): Non-parametric learning method that creates a model predicting values by learning simple decision rules inferred from features.

AdaBoost: An ensemble method that combines multiple "weak" learners (shallow decision trees) into a strong learner by iteratively focusing on misclassified samples.

Support Vector Machines (SVM): Optimization of the margin between classes. We test three kernels:
- *Linear:* Standard dot-product. Ideal for linearly separable data.
- *Polynomial (d=3):* Standard polynomial expansion.
- *RBF (Gaussian):* Measures local similarity via radial basis functions.

3. Experimental Studies

3.1 Data Distribution

The 3D Make-Moons data is generated with a noise level of 0.2. As shown in Figure 1, the two classes are highly non-linear and exhibit a manifold-like structure in 3D space.

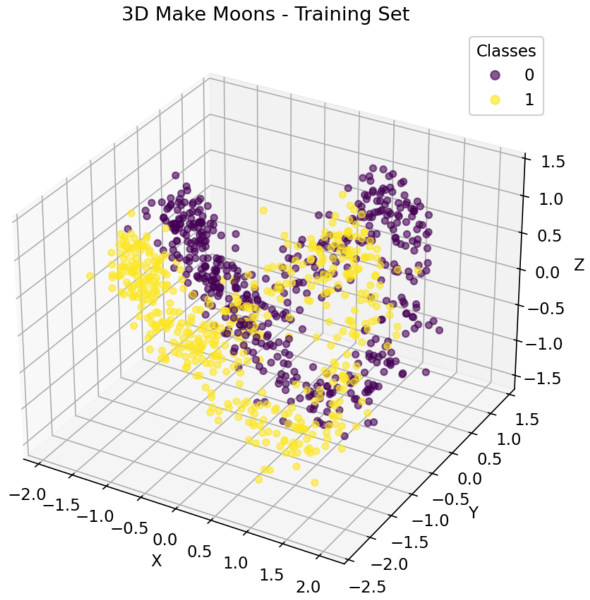

Figure 1: 3D Make-Moons training data distribution showing the non-linear manifold.

3.2 Quantitative Results

Table 1 summarizes the accuracy and F1-score for all models on the held-out test set (500 samples).

Model	Accuracy	F1-Score	Grade
Decision Tree	0.9540	0.9533	Good
AdaBoost (DT)	0.9900	0.9900	Best
SVM (Linear)	0.6980	0.7010	Poor
SVM (Poly d=3)	0.8800	0.8661	Moderate
SVM (RBF)	0.9880	0.9880	Excellent

Table 1: Classification results comparison on 3D test set.

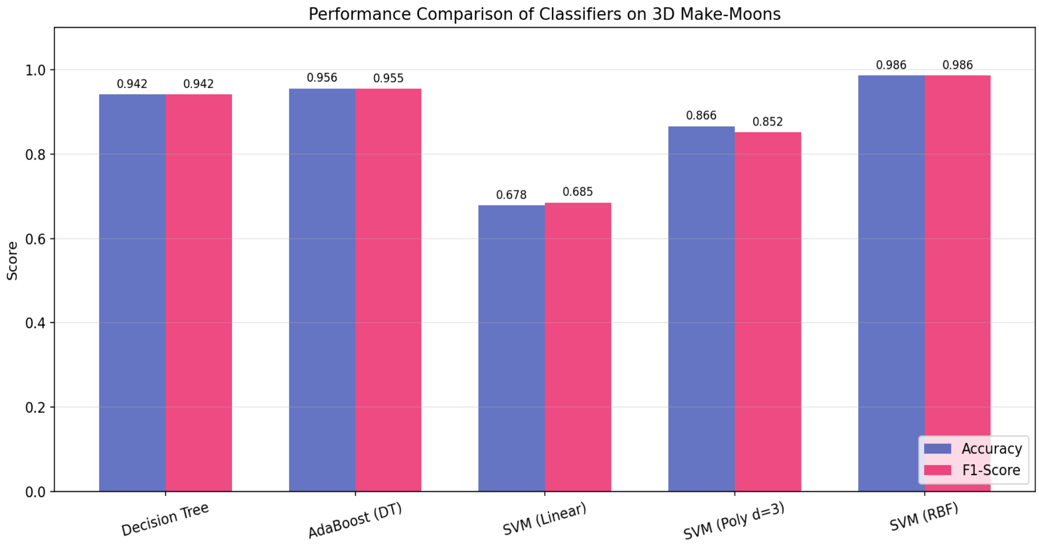

Figure 2: Performance metrics comparison across all five models.

4. Conclusions

The experiments conclude that:

(1) Linear SVM is unsuitable for Moons-pattern data, as no simple hyper-plane can separate the classes.

(2) SVM with RBF kernel and the AdaBoost ensemble successfully captured the complex data structure.

(3) Ensemble learning (AdaBoost) proved most effective, reaching 99% accuracy through iterative boundary refinement.

References

[1] C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
[2] F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR, 2011.